SHAHABUDEEN NS

Embedded and VLSI Engineer

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PROFESSIONAL SUMMARY

A motivated Electronics and Communication Engineering graduate with strong fundamentals in core electronics and analytical problem-solving skills. Proficient in PCB and VLSI design, with practical exposure to design flow, layout, and optimization.Experienced in embedded systems, with hands-on skills in microcontroller-based development and hardware interfacing.

PROJECTS

Smart EV Charging Monitoring System

06/2026 – 07/2026

- Developed an ESP32-based IoT system to monitor voltage, current, power, energy, and charging cost.
- Integrated Firebase, ThingSpeak, and GSM (SIM800L) for cloud monitoring and SMS alerts.
- Designed the complete PCB schematic and layout using KiCad. and interfaced ZMPT101B, ACS712, and I^2C LCD using Embedded C.

Railway Track Crack Detection System

01/2026 – 03/2026

- Developed an ESP32-based system for detecting railway track cracks using vibration and pressure sensors.
- Integrated LoRa communication and IoT for real-time monitoring and instant fault alerts.
- Improved detection reliability through multi-sensor validation and instant alert generation.

Low-Voltage AC Overhead Conductor Breakage Detection System

08/2025 – 09/2025

- Developed an ESP32-based system for detecting breakage in low-voltage AC overhead conductors using voltage and current sensing.
- Implemented TinyML for real-time fault detection with low latency.
- Enabled IoT-based monitoring and instant fault alerts for improved reliability.

Object Detection and Monitoring System for Conveyor Belt Using ToF Sensors

08/2025

- Developed an Arduino-based system using Time-of-Flight sensors for real-time object detection on a conveyor belt.
- Enabled accurate monitoring and control to improve automation efficiency and operational reliability.

EDUCATION

B.E. in Electronics and Communication Engineering

Government College of Engineering, Salem

2023 – 2027

CGPA : 8.03 / 10.00 (Upto 6th semester)

HSC

SRV Boys Higher Secondary School, Rasipuram

2022 – 2023

Percentage : 93.5%

TECHNICAL SKILLS

Programming Languages

- C
- C++
- Python
- Java
- Embedded C
- Verilog

Core Skills

- Core Electronics and Circuit Analysis
- PCB Design
- Embedded Systems (ESP32, STM32, Arduino)
- Communication Protocols(UART,I2C,SPI)

EDA & Development Tools

- KiCad
- Cadence Virtuoso
- Vivado
- STM32CubeIDE
- Arduino IDE

ACHIEVEMENTS

Smart India Hackathon (SIH) - 2025

08/2025 – 09/2025

Selected in the internal round of the **Smart India Hackathon** for proposing an embedded and intelligent monitoring solution to detect breakage in low-voltage AC overhead distribution conductors.

IIT BOMBAY FOSSEE - AAI GEOSPATIAL MAPATHON 2025

04/2025 – 05/2025

Recognized as **Winner** at the **IIT Bombay FOSSEE – AAI Geospatial Mapathon 2025** for developing a waste management analysis solution using geospatial mapping, spatial data analysis, and visualization techniques at the Chennai dumpyard.

WORKSHOPS

Voice of the Future

Workshop based on Speech Recognition and Computer Vision

03/2025 | PSG College of Technology

Embedded System with IOT

26/03/2025

Hands-on Training on Measuring Instruments

22/04/2025

INTERNSHIP

Tamirabot Advanced Engineering Pvt. Ltd.

06/2026 – 07/2026

- Learned the fundamentals of EV charging systems and their monitoring techniques.
- Gained hands-on experience in 3D printing for prototype and enclosure development.
- Assisted in embedded hardware development and system testing.